

SEG800 DSP Cheatsheet

Test 3: z-Transform, DTFT, Frequency-Domain LTI Analysis, Sampling

1 Test Attack

1. Prove periodic

$$x[n + N] = x[n], \quad N \in \mathbb{Z}^+$$

For

$$x[n] = A \cos(\omega_0 n + \theta)$$

periodic iff

$$\omega_0 N = 2\pi k \quad \Longleftrightarrow \quad \frac{\omega_0}{2\pi} = \frac{k}{N}$$

If k/N is in lowest terms, the fundamental period is N . Same idea for $e^{j\omega_0 n}$.

2. Find DTFT / inverse DTFT

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

Try in this order: standard pair, geometric series, time shift, differentiation trick.

3. Given a system, find magnitude and phase

$$Y(\omega) = X(\omega)H(\omega)$$

Find $H(\omega)$ from $h[n]$ or the difference equation, then compute $|H(\omega)|$ and $\angle H(\omega)$.

2 Part I: The Z-Transform

Definition:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \quad x[n] \xleftrightarrow{\mathcal{Z}} X(z)$$

Complex variable: $z = re^{j\omega}$.

2.1 ROC Rules

ROC = set of z where the sum converges.

- ROC never contains a pole.
- Right-sided: outside outermost pole.
- Left-sided: inside innermost pole.
- Two-sided: between poles.
- Stable: ROC includes $|z| = 1$.
- Causal rational stable: all uncanceled poles inside unit circle.

2.2 Core Pairs

$$\begin{aligned} \delta[n] &\xleftrightarrow{\mathcal{Z}} 1 \\ u[n] &\xleftrightarrow{\mathcal{Z}} \frac{1}{1 - z^{-1}}, \quad |z| > 1 \\ a^n u[n] &\xleftrightarrow{\mathcal{Z}} \frac{1}{1 - az^{-1}}, \quad |z| > |a| \\ na^n u[n] &\xleftrightarrow{\mathcal{Z}} \frac{az^{-1}}{(1 - az^{-1})^2}, \quad |z| > |a| \\ -a^n u[-n-1] &\xleftrightarrow{\mathcal{Z}} \frac{1}{1 - az^{-1}}, \quad |z| < |a| \\ -na^n u[-n-1] &\xleftrightarrow{\mathcal{Z}} \frac{az^{-1}}{(1 - az^{-1})^2}, \quad |z| < |a| \\ (\cos \omega_0 n) u[n] &\xleftrightarrow{\mathcal{Z}} \frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}, \quad |z| > 1 \\ (\sin \omega_0 n) u[n] &\xleftrightarrow{\mathcal{Z}} \frac{z^{-1} \sin \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}, \quad |z| > 1 \end{aligned}$$

Damped trig: replace denominator by $1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}$ and ROC by $|z| > |a|$.

2.3 Key Properties

$$\begin{aligned} a_1 x_1[n] + a_2 x_2[n] &\xleftrightarrow{\mathcal{Z}} a_1 X_1(z) + a_2 X_2(z) \\ x[n-k] &\xleftrightarrow{\mathcal{Z}} z^{-k} X(z), \quad x[-n] \xleftrightarrow{\mathcal{Z}} X(z^{-1}) \\ a^n x[n] &\xleftrightarrow{\mathcal{Z}} X(a^{-1} z) \\ x_1[n] * x_2[n] &\xleftrightarrow{\mathcal{Z}} X_1(z) X_2(z) \\ x[0] &= \lim_{z \rightarrow \infty} X(z) \quad (\text{causal}) \end{aligned}$$

2.4 Inverse z-Transform

Methods: power series and partial fractions.

$$\frac{1}{1-a} = \sum_{n=0}^{\infty} a^n, \quad |a| < 1$$

Power-series idea:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Read off the coefficients of z^{-n} . Outside-pole ROC $\Rightarrow$ expand in powers of z^{-1} .

$$\mathcal{Z}^{-1} \left\{ \frac{1}{1 - pz^{-1}} \right\} = \begin{cases} p^n u[n], & |z| > |p| \\ -p^n u[-n-1], & |z| < |p| \end{cases}$$

Example:

$$\frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}} = \frac{2}{1 - z^{-1}} - \frac{1}{1 - 0.5z^{-1}}$$
$$|z| > 1 \Rightarrow x[n] = \left(2 - \left(\frac{1}{2}\right)^n\right) u[n]$$

2.5 System Function

For a relaxed LTI system,

$$y[n] = -\sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

3 Part II: DTFT

3.1 Core Formulas

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

DTFT is continuous in ω and periodic with period 2π .

$$X(\omega + 2\pi) = \sum x[n] e^{-j(\omega+2\pi)n} = X(\omega)$$

Use $-\pi \leq \omega \leq \pi$ as the main interval.

3.2 Helpful CTFT Reminder

Continuous-time nonperiodic FT:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt, \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Pairs: $\delta(t) \xleftrightarrow{\mathcal{F}} 1$, $e^{-at} u(t) \xleftrightarrow{\mathcal{F}} 1/(a + j\omega)$, time pulse $\leftrightarrow$ sinc.

3.3 Must-Know DTFT Pairs

$$\begin{aligned} \delta[n] &\xleftrightarrow{\mathcal{F}} 1 \\ 1 &\xleftrightarrow{\mathcal{F}} 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k) \\ a^n u[n] &\xleftrightarrow{\mathcal{F}} \frac{1}{1 - ae^{-j\omega}}, \quad |a| < 1 \\ na^n u[n] &\xleftrightarrow{\mathcal{F}} \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2}, \quad |a| < 1 \end{aligned}$$

Ideal discrete-time lowpass rectangle $\leftrightarrow$ sinc sequence: $x[0] = \omega_c/\pi$, $x[n] = \sin(\omega_c n)/(\pi n)$ for $n \neq 0$.

3.4 Standard Problem Templates

Finite rectangular sequence, $x[n] = 1$ for $0 \leq n \leq N-1$:

$$X(\omega) = \sum_{n=0}^{N-1} e^{-j\omega n} = e^{-j\omega(N-1)/2} \frac{\sin(N\omega/2)}{\sin(\omega/2)}$$

For $N = 4$:

$$X(\omega) = e^{-j3\omega/2} \frac{\sin(2\omega)}{\sin(\omega/2)}$$

$$|X(\omega)| = \left| \frac{\sin(2\omega)}{\sin(\omega/2)} \right|$$

phase = $-3\omega/2$ with possible π jumps when the real ratio is negative.

3.5 Properties You Actually Use

$$x[n-k] \xleftrightarrow{\mathcal{F}} e^{-j\omega k} X(\omega), \quad x[-n] \xleftrightarrow{\mathcal{F}} X(-\omega)$$

$$x_1[n] * x_2[n] \xleftrightarrow{\mathcal{F}} X_1(\omega) X_2(\omega)$$

$$e^{j\omega_0 n} x[n] \xleftrightarrow{\mathcal{F}} X(\omega - \omega_0)$$

$$nx[n] \xleftrightarrow{\mathcal{F}} j \frac{dX(\omega)}{d\omega}$$

Example:

$$n(0.5)^n u[n] \xleftrightarrow{\mathcal{F}} \frac{0.5e^{-j\omega}}{(1 - 0.5e^{-j\omega})^2}$$

3.6 Symmetry and Parseval

If $x[n]$ is real,

$$X(\omega) = X^*(-\omega)$$

So $X_R(\omega)$ and $|X(\omega)|$ are even; $X_I(\omega)$ and phase are odd. Real even $x[n] \Rightarrow$ real even $X(\omega)$.

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$$

4 Part III: LTI Frequency Analysis and Sampling

4.1 Core LTI Frequency Formulas

$$y[n] = x[n] * h[n] \quad \Longleftrightarrow \quad Y(\omega) = X(\omega)H(\omega)$$

$$H(\omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}, \quad h[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega)e^{j\omega k} d\omega$$

$H(\omega)$ is 2π -periodic.

4.2 Professor Hint 3: Magnitude and Phase

If

$$H(\omega) = H_R(\omega) + jH_I(\omega) = |H(\omega)|e^{j\Theta(\omega)}$$

then

$$|H(\omega)| = \sqrt{H_R^2(\omega) + H_I^2(\omega)}$$

$$\Theta(\omega) = \angle H(\omega) = \tan^{-1} \left(\frac{H_I(\omega)}{H_R(\omega)} \right)$$

If

$$H(\omega) = \frac{N(\omega)}{D(\omega)}$$

then

$$|H(\omega)| = \frac{|N(\omega)|}{|D(\omega)|}, \quad \angle H(\omega) = \angle N(\omega) - \angle D(\omega)$$

Also,

$$|Y(\omega)| = |X(\omega)| |H(\omega)|, \quad \angle Y(\omega) = \angle X(\omega) + \angle H(\omega)$$

For delay factor $e^{-j\omega k}$:

$$|e^{-j\omega k}| = 1, \quad \angle e^{-j\omega k} = -\omega k$$

4.3 Eigenfunction Shortcut

If the input is

$$x[n] = Ae^{j\omega_0 n}$$

then

$$y[n] = AH(\omega_0)e^{j\omega_0 n}$$

For a real sinusoid,

$$y[n] = A|H(\omega_0)| \cos(\omega_0 n + \phi + \angle H(\omega_0))$$

4.4 Find $H(\omega)$ from a Difference Equation

Let

$$x[n] = e^{j\omega n}, \quad y[n] = H(\omega)e^{j\omega n}$$

Substitute and divide by $e^{j\omega n}$. Example:

$$y[n] + 0.5y[n-1] + 0.25y[n-2] = x[n] + 2x[n-1] + x[n-2]$$

gives

$$H(\omega) = \frac{1 + 2e^{-j\omega} + e^{-j2\omega}}{1 + 0.5e^{-j\omega} + 0.25e^{-j2\omega}}$$

4.5 First-Order Lowpass / Smoother

For

$$y[n] = ay[n-1] + bx[n], \quad 0 < a < 1$$

$$H(\omega) = \frac{b}{1 - ae^{-j\omega}}$$

with

$$1 - ae^{-j\omega} = (1 - a \cos \omega) + ja \sin \omega$$

So

$$|H(\omega)| = \frac{|b|}{\sqrt{(1 - a \cos \omega)^2 + (a \sin \omega)^2}} = \frac{|b|}{\sqrt{1 + a^2 - 2a \cos \omega}}$$

$$\angle H(\omega) = \angle b - \tan^{-1} \left(\frac{a \sin \omega}{1 - a \cos \omega} \right)$$

Maximum gain occurs at $\omega = 0$:

$$|H(0)| = \frac{|b|}{1 - a}$$

Choose

$$b = 1 - a$$

to make the maximum magnitude unity. Then

$$H(\omega) = \frac{1 - a}{1 - ae^{-j\omega}}$$

is a normalized lowpass filter.

4.6 Complex-Exponential Output Example

If

$$h[n] = \left(\frac{1}{2}\right)^n u[n], \quad H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

For $x[n] = Ae^{j\pi n/2}$,

$$H\left(\frac{\pi}{2}\right) = \frac{1}{1 + \frac{j}{2}} = \frac{2}{\sqrt{5}}e^{-j26.6^\circ}$$

$$y[n] = \frac{2A}{\sqrt{5}}e^{j(\pi n/2 - 26.6^\circ)}$$

For $x[n] = Ae^{j\pi n}$,

$$H(\pi) = \frac{2}{3}, \quad y[n] = \frac{2}{3}Ae^{j\pi n}$$

4.7 Sampling

$$x[n] = x_a(nT), \quad F_s = \frac{1}{T}$$

For $x_a(t) = A \cos(2\pi Ft + \theta)$,

$$x[n] = A \cos\left(2\pi \frac{F}{F_s}n + \theta\right)$$

Frequency conversions:

$$\Omega = 2\pi F, \quad f = \frac{F}{F_s}, \quad \omega = 2\pi f = 2\pi \frac{F}{F_s} = \Omega T$$

$$F = fF_s = \frac{\omega F_s}{2\pi}, \quad \Omega = \omega F_s$$

Because $e^{j(\omega + 2\pi k)n} = e^{j\omega n}$, discrete-time frequency is periodic with period 2π .

4.8 Aliasing and Nyquist

If

$$F_2 = F_1 + kF_s$$

then both analog frequencies produce the same sampled complex exponential. Fold into

$$-\frac{F_s}{2} \leq F \leq \frac{F_s}{2}$$

using

$$F_{\text{alias}} = F - kF_s$$

Nyquist frequency:

$$F_{\text{Nyquist}} = \frac{F_s}{2}$$

Sampling theorem for bandlimit B :

$$F_s \geq 2B$$

Nyquist rate = $2B$. Reconstruction uses sinc interpolation:

$$x_a(t) = \sum_{n=-\infty}^{\infty} x_a\left(\frac{n}{F_s}\right) g\left(t - \frac{n}{F_s}\right), \quad g(t) = \frac{\sin(2\pi Bt)}{2\pi Bt}$$